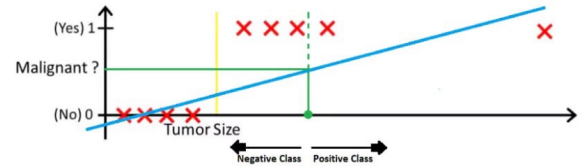
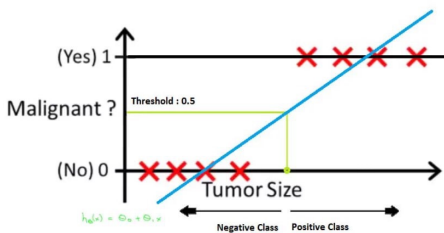


Defn:- Logistic regression is a supervised learning algorithm used for binary classification problems.

we use it when the target variable is categorical with two outcomes.

- Despite the regression in its name, it is used to predict the probability that an observation belongs to a particular category.
- It models the probability using the sigmoid function, which maps a linear combination of input features to a value between 0 and 1.
- why can't we use Linear Regression for a classification problem?

-
1. **Unbounded Output:** Linear regression produces continuous values that can be outside the $[0, 1]$ range, which cannot be interpreted as probabilities.
 2. **Sensitivity to Outliers:** The best-fit line in linear regression is highly sensitive to outliers. A single outlier in a classification context can drastically shift the decision boundary, leading to poor performance.



Sigmoid function | Logistic Regression

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid function, defined as $\sigma(z) = 1 / (1 + e^{-z})$, is crucial because it transforms any real-valued number from a linear combination of features (z) into a value between 0 and 1. This output is interpreted as a probability. A result close to 1 indicates a high probability of belonging to Class 1, while a value close to 0 indicates a high probability of belonging to Class 0.

Logistic Regression Model

$$\text{Let } z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

$$\begin{aligned} P(y_i = 1 | \underline{x}) &= \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}} \\ &= \sigma(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n) = \sigma(z) \end{aligned}$$

Odds: The odds are the ratio of something happening to something not happening.

Ex: odds in favour of my team winning the game 1 to 4
odds (in favour) = $\frac{1}{4}$

" " " " " " " " " 5 to 3
odds (in favour) = $\frac{5}{3}$

Note: odds are not probabilities.

- If an event has probability p of happening, the odds (in favour) = $\frac{p}{1-p}$.

- Odds in favour of event E are $m:n$

$$\Rightarrow P(E) = \frac{m}{m+n}$$

Odds against event E are $m:n$

$$\Rightarrow P(E^c) = \frac{m}{m+n}.$$

Let $p = P(Y=1 | \underline{x})$

$$\text{odds} = \frac{p}{1-p}$$

Log-odds (Logit):

$$\text{Logit}(p) = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

- The model assumes linear relationship between the features (x_i) and log-odds of the event.
- In logistic regression, we model the logit | log-odds as a linear function of features. This is why the model is called "logistic regression".

$$\therefore p = \frac{1}{1 + e^{-\ln(\text{Odds})}}$$

Training:- The model is trained using Maximum Likelihood Estimation (MLE). Optimization algorithms like Gradient Descent are used to solve maximization problem.

Loss function :-

$$\begin{aligned} &\text{log loss or Binary cross entropy} \\ &= - \left\{ y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right\} \end{aligned}$$

Assumptions of Logistic Regression :

- i) The dependent variable must be binary.
- ii) The data points should be independent of each other.
- iii) There should be a linear relationship between the independent variables and log-odds of the outcome.
- iv) Independent variables shouldn't be highly correlated with each other.
- v) Large sample size is generally beneficial for stable estimates.

Interpretation of coefficients :-

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

- In logistic regression, coefficients represent the change in log-odds of the outcome for a one-unit change in predictors variable, holding all other variables constant.

Example Interpretation:

If we have a model: $\log(\text{odds of default}) = -3 + 0.2 * (\text{income}) - 0.5 * (\text{debt_ratio})$

- **Intercept ($\beta_0 = -3$):** When all predictors are zero, the log-odds of default is -3
- **Income coefficient ($\beta_1 = 0.2$):** For each \$1,000 increase in income, the log-odds of default increases by 0.2
- **Debt ratio coefficient ($\beta_2 = -0.5$):** For each unit increase in debt-to-income ratio, the log-odds of default decreases by 0.5

Odds ratio:- An odds ratio (OR) compares odds between two groups.

$$OR = \frac{\text{Odds group A}}{\text{Odds group B}}$$

Odds Ratio Interpretation (More Intuitive)

$$OR = e^{\beta}$$

Interpretation Guidelines:

- $OR = 1$: No effect ($\beta = 0$)
- $OR > 1$: Positive relationship with outcome
- $OR < 1$: Negative relationship with outcome

Continuing our example:

- $OR_{\text{income}} = e^{0.2} = 1.22$
- $OR_{\text{debt_ratio}} = e^{(-0.5)} = 0.61$

Income: For each \$1000 increase in income the odds of default increase by 22%.
(multiply by 1.22)

Debt_ratio: For each unit increase in debt ratio, the odds of default decreased by 39%.
(multiply by 0.61)

Binary/Categorical Variables:

Scenario: "Home_ownership" (1 = homeowner, 0 = renter)

- Coefficient: -0.8
- Odds Ratio: $e^{(-0.8)} = 0.45$
- Interpretation: Homeowners have 55% lower odds of default compared to renters

Ordinal Variables:

Scenario: "Employment_length" (0: <1 year, 1: 1-3 years, 2: 3-5 years, 3: >5 years)

- Coefficient: -0.4
- Odds Ratio: $e^{(-0.4)} = 0.67$
- Interpretation: For each category increase in employment length, the odds of default decrease by 33%



Example: Categorical Predictor with More Than Two Levels

Now, let's look at a categorical predictor with multiple levels, such as `InternetService` with the categories: "None" (reference category), "DSL", and "Fiber optic".

Imagine the model provides the following coefficients:

- `InternetService_DSL` : 0.92
- `InternetService_Fiber optic` : 1.25

Here's how to interpret these:

- Interpretation for DSL:
 - Log-Odds: Customers with DSL have 0.92 higher log-odds of churning compared to customers with no internet service.
 - Odds Ratio: $e^{0.92} \approx 2.51$. The odds of a customer with DSL churning are 2.51 times the odds of a customer with no internet service churning.
- Interpretation for Fiber Optic:
 - Log-Odds: Customers with Fiber optic have 1.25 higher log-odds of churning compared to customers with no internet service.
 - Odds Ratio: $e^{1.25} \approx 3.49$. The odds of a customer with Fiber optic service churning are 3.49 times the odds of a customer with no internet service churning.

Comparing Non-Reference Categories: To compare DSL and Fiber optic, you can look at the difference in their coefficients and the ratio of their odds ratios. Since the coefficient for Fiber optic (1.25) is larger than for DSL (0.92), we can conclude that customers with Fiber optic have a higher likelihood of churning than those with DSL.







